

Smart CCTV Using Computer Vision: A Case Study of Bangladesh

Shekh Muhammad Tanveer Rahman^{1*}, Xiulan Hao^{1*,2*}

¹School of Information Engineering, Huzhou University, Hu'zhou, 313000, Zhejiang Province, China

² Zhejiang Province Key Laboratory of Smart Management & Application of Modern Agricultural Resources, Huzhou University, Hu'zhou, 313000, Zhejiang Province, China)

Abstract: Public security, such as running, strolling, and jumping on roads, and theft, is a Challenging problem in Bangladesh. How can Computer Vision improve Security Challenges in Bangladesh? Specifically, what does the current security system in Bangladesh look like, what main points occur during the exam monitoring, and what does the proposed computer vision solution look like? Our solution is to use Closed Circuit cameras (CCTV) to detect changes in the scene and record video clips that have changes in retrospect of Bangladesh. We attempted a strategy for spotting abnormalities in the supplied footage and instantly notifying the appropriate authorities by email. It uses neural networks and convolutional neural networks to record videos where some change has happened and tag those frames with respect to what activity a human or a group of humans were performing. Primary Data is collected by Extensive research of the existing articles on the subject matter and their corresponding applications. Secondary data is collected by survey, interviews. The data was separated into training and testing in an 80:20 ratio in the first stage, and the foreground data was obtained by subtracting the background data. The model was created, and feature extraction was performed using an auto-encoder architecture. Lastly, aberrant occurrences are discovered using pattern matching. We utilized the self-imported dataset, which is 7 GB in size. The suggested approach has a 5-epoch accuracy of 54.47%, a 15-epoch accuracy of 64.30% and a 50-epoch accuracy of 80.76%.

Keywords: Closed Circuit cameras (CCTV), Bangladesh; Computer Vision, Pattern Matching, YOLO,

*Corresponding Author

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Introduction

At first let us imagine a scenario of a robbery, where some members of the gang, are trying to remove the door while their faces are covered. What usually happens in Bangladesh is that the CCTV cameras record everything, but it is not able to understand the actual scenario. And it has no way of judging whether the scene is changing or not. If there were a system which can detect the activities inside every CCTV camera's view that would be very helpful for the security purposes. If cameras can be used to detect and alert authorities when there is a change or activity in front of a CCTV camera, allowing them to be sent to the owner or authorities more easily than viewing and analyzing all videos. The most important details presented in this paper are that CCTV cameras will be used to detect changes in the scene and

record the video clips that have changes in retrospect of Bangladesh. This will make the CCTV much smarter with respect to storage, and the same storage module can be used to send alerts to the owner if needed. Additionally, pointers will be given on how to do that if the change happens due to human activities. People have been concerned about their security since the birth of the modern age. Every day millions are spent in this sector. But the developing and underdeveloped country tends to give less attention to this area. Every year, a large amount of the population reconciles gun-related violence all over the world. This system can also be implemented in high-end surveillance and security systems. It can detect a weapon or unsafe assets to avoid any kind of assault or risk to human life [16]. Furthermore, even though due to COVID-19, exams have been occurring online, offline examinations have yet to lose their charm. Cheating in the exam has been occurring since the very beginning of the testing system. If there is a smart system to detect the cheaters, the hard-working people will get their recognition.

Related Works

In his work [6] wrote that the System consists of developing a smart tool by which a computer or Raspberry Pi attached with a camera can be converted into a smart surveillance system. This project aims to develop a smart and cost-efficient smart system for home and business owners and small business owners. In another work [11] Automatic License Plate Recognition (ALPR) has been a frequent topic of research. This paper presents a robust and efficient ALPR system based on the state-of-the-art YOLO object detector. Convolutional Neural Networks are trained and fine-tuned for each ALPR stage. In an article by [14], they have analyzed a system to regulate traffic congestion in major cities. The system is based on the traffic density that must be calculated in each lane. This will help to reduce traffic congestion, road accidents, and energy consumption and waiting time. There are even research concerning a smart city where detection is done via CCTV. The development of cities that embrace the concept of a smart city is occurring all over the world. Since the year 2000, smart cities have been implemented in the city of Jakarta. The system has been used in almost every aspect of government, from absences to public service applications [15]. In another detection using YOLO [7] stated that their research proposes to transform real-time unstructured data from CCTV into knowledge that can be played in a web dashboard. YOLO framework with YoloV4 Architecture and COCO dataset for traffic flow counting and illegal parking detection. The prototype result can detect the object at 10 frames per second. The research was also present regarding the surveillance system which uses CNN. They wrote [9] their study aims to construct a real-world surveillance system that can effectively detect moving persons using limited resources. New datasets are constructed by filming alleys and playgrounds, places where crimes are likely to occur. The best classification accuracy of 0.85 was obtained for a test set from the same camera with a training set and different cameras.

Significance and Objective

Research Gap

CCTV or Closed-Circuit Television Cameras are mainly dumb; they only can record real-time motions as a video. Nowadays several systems added into the surveillance system which includes detection of objects, Thermal scanning, Motion detection etc. This research aims to develop a system prototype that can automatically detect activities based on actions in a Video. The existing literature and system reference a desktop version of smart security systems. There has been little quantitative analysis of its mobile functionality. Bangladesh barely use CCTV to organize their securities. Even though one can find CCTV cameras in every nook and corner of the country the security issues detected by those devices consume a huge amount of time and manual labor. This paper evaluates the application of object detection in moving scenarios and aims to generate a prototype. It offers preliminary insights on computer vision's application in the security regions and explores the south-east Asian challenges. The project provides an opportunity to advance the understanding of the security challenges of Bangladesh and presents opportunities for further improvement of the system. An Automated system for object detection would lead to greater help for all the security sectors in Bangladesh. This system can help the invigilators after exams to detect cheating in a very short amount of time. The proposed smart CCTV prototype can not only save time but also reduce the manual labor of searching through trillions of visual data.

Bangladesh is a developing country and day by day it is facing newly developed security threats. Only recently the country has started using CCTV in the national range. But it needs a lot of data monitoring. **If a smart system is developed it will not only save time but also will help increase security.** Furthermore, there is no proper way to detect cheating during offline exams. It can be assumed that an installation of a user-friendly system to ensure process simplification would curb the current challenges.



Fig. 1: CCTV installed in Bangladesh.

Research Objectives

The main objective of the research is to develop a design model that implements the concepts of computer vision to improve the security issues for Bangladesh. Specifically, this study aims at achieving the following: to investigate the current security systems of Bangladesh, to identify the key pain points as well as relevant computer vision features that can be implemented, to propose a new computer vision-based solution to tackle identified pain points, to test the application of a smart CCTV system in the computer vision architecture.

Computer Vision

The study of replicating human vision processes, passing them on to machines, and automating them is known as computer vision [20]. **It is a branch of artificial intelligence that teaches computers to evaluate visual environments.** It is the process of extracting information, comprehending it, and learning from a single image. It is the process of a machine receiving and analyzing visual data to enable a computer to perform tasks with the same accuracy as humans. As a result, **it has a close relationship with machine learning and image processing.** **The image can take many forms, including video 13 sequences, depth images, multiple camera views, or multi-dimensional data from a medical scanner [8].** When **computer vision first started in the early 1970s**, it was viewed as the visual perception component of an ambitious agenda to mimic human intelligence and to endow robots with intelligent behavior [13]. If we look at modern-day implementation, we can say that computer vision has become an important part of the technological community. Now the main question is how can computer vision help in our smart CCTV project? If we look at the following flowchart, we can get a little overview of how computer vision can enhance our system:

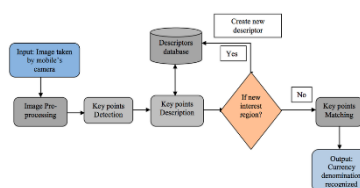


Fig. 2: Computer Vision in context of object detection [5]

Design and Methodology

Image Transformation

In image transformations several methods are worth mentioning. **The most important one, which is related to this study is Gray Level Histogram Method.** Generally, given a gray level histogram from an image as below. **Changing the histogram to uniform distribution from an image is usually what we called Histogram equalization [28].**

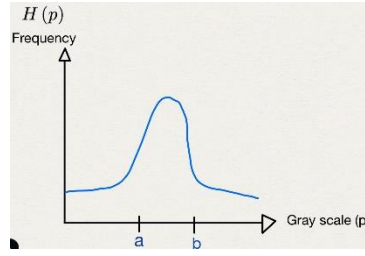


Fig. 3: Graphical representation of Gray Level Histogram [30]

In discrete time, the area of gray level histogram is,

$$\sum_{i=0}^k H(p_i) \quad (1)$$

while the area of uniform distribution is

$$\sum_{i=0}^k G(q_i) \quad (2)$$

It's clear that the area won't change, so

$$\sum_{i=0}^k H(p_i) = \sum_{i=0}^k G(q_i) \quad (3)$$

From the uniform distribution, the probability of

$$q_i = \frac{N^2}{q_k - q_0} \quad (4)$$

while the $0 < i < k$ in continuous time, the equation is.

$$\int_{q_0}^q \frac{N^2}{q_k - q_0} ds = \int_{p_0}^p H(s) ds \quad (5)$$

Moreover, based on the definition of a function, the gray level histogram method is like finding a function f that satisfies,

$$f(p)=q \quad (6)$$

Feature Extraction [26]: Two methods are mainly used for image extraction from a video or image. discussed here a) To extract features of a face at first the image is converted into a binary. From this binary image the centroid (X, Y) of the face image is calculated using equation 7 and 8,

$$X = \frac{\sum mx}{\sum x} \quad (7)$$

$$Y = \frac{\sum my}{\sum y} \quad (8)$$

Algorithms

This system will be using a variety of neural networks. A neural network is a network or circuit of neurons, or in a modern sense, an artificial neural network, composed of artificial neurons or nodes [19]. Neural Networks use a basis function that follows the same form as Linear models for regression and classification. The basic neural network model, which can be described as a series of functional transformations. First M linear combinations are constructed of the input variables $x_1, \dots, x_D$ in the form.

$$a_j = \sum_{i=1}^D w_{ji}^{(1)} x_i + w_{j0}^{(1)} \quad (9)$$

where $j=1 \dots M$, and the superscript (1) indicates that the corresponding parameters are in the first 'layer' of the network. The parameters $w_{ji}^{(1)}$ is referred to as weights and the parameters $w_{j0}^{(1)}$ as biases. The quantities a_j are known as activations. [3].

CNN

Convolutional Neural Network or Commonly known as CNN is commonly used for visual image analysis. In other words, Convolutional neural networks are deep learning algorithms that can train large datasets with millions of

parameters as input in the form of 2D images and convolve them with filters to produce the desired outputs [4].

Consider the task of recognizing handwritten digits. The identity of the digit is invariant under translations and scaling as well as (small) rotations [21]. A simple approach would be to treat the image as the input to a fully connected network. This ignores a key property of images, which is that nearby pixels are more strongly correlated than distant pixels [3].

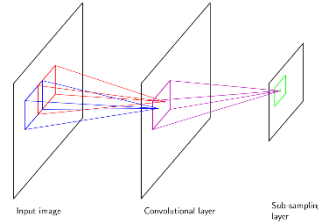


Fig. 4: Diagram illustrating part of a convolutional neural network, showing a layer of convolutional units followed by a layer of subsampling units. Several successive pairs of such layers may be used [3].

DCNN

A deep convolutional neural network (DCNN) consists of many neural network layers. Two different types of layers, convolutional and pooling, are typically alternated. The depth of each filter increases from left to right in the network [12].

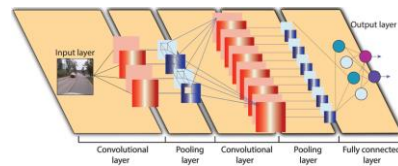


Fig. 5: A general DCNN architecture [12].

We consider real-world image blur that suffers from several types of degradation including clipped intensity (saturation), camera noise, and compression artifacts. The blur model is given by,

$$\hat{y} = \psi b[\varphi(\alpha x * k + n)] \quad (10)$$

where αx represents the latent sharp image. The notation $\alpha \geq 1$ is to indicate the fact that αx could have values exceeding the dynamic range of camera sensors and thus be clipped. k is the known convolution kernel, or typically referred to as a point spread function (PSF), n models additive camera noise. $\varphi(\cdot)$ is a clipping function to model saturation, defined as $\varphi(z) = \min(z, z_{\max})$, where $z_{\max}$ is a range threshold. $\psi b[\cdot]$ is a nonlinear (e.g., JPEG) compression operator.

We note that even with $\hat{y}$ and kernel k , restoring αx is intractable, simply because the information loss caused by clipping. In this regard, our goal is to restore the clipped input $\hat{x}$, where $\hat{x} = \varphi(\alpha x)$.

Although solving for $\hat{x}$ with a complex energy function that involves Eq. (1) is difficult, the generation of blurry image from an input x is quite straightforward by image synthesis according to the convolution model taking all kinds of possible image degradation into generation. This motivates a learning procedure for deconvolution, using training image pairs $\{\hat{x}_i, \hat{y}_i\}$, where index $i \in N$.

The deconvolution task can be approximated by a convolutional network by nature. We consider the following simple linear blur model [29].

$$y = x * k \quad (11)$$

The spatial convolution can be transformed to a frequency domain multiplication, yielding.

$$F(y) = F(x) \cdot F(k) \quad (12)$$

YOLO

Commonly known as YOLO or “You Only Look Once” is an algorithm that uses neural networks to provide real-time object detection. This algorithm is popular because of its speed and accuracy. It has been used in various applications to detect traffic signals, people, parking meters, and animals [18]. In this prototype, YOLO is used for object detection especially a sequence of images.

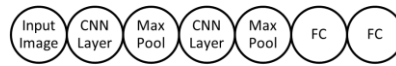


Fig. 6: Input Image Processing using YOLO [22]

Research Design

An entire process mapping and modelling exercise is done to describe and analyze the current processes. Primary data is collected by extensive research of the subject matter and its corresponding applications. Secondary data is collected through survey, interviews, article reading, and test running existing systems to find any gaps. This prototype will aim to do so. Survey research is an important part of research, used to gather information from individuals and groups. It can range from targeted questions to more rigorous studies, such as marketing or political surveys [27]. Over the span of 2 years, I have conducted several surveys to learn about people’s preference. The very first survey that I have conducted was to determine the status of security in Bangladesh.

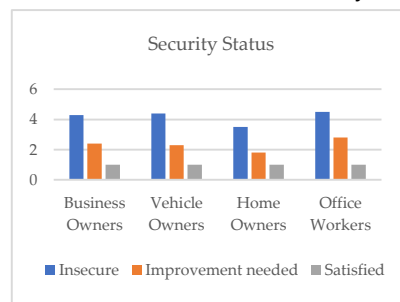


Fig 7: Security Status of Bangladesh

People in Bangladesh feel insecure and want improvement in their security systems, as evidenced by the results of a survey asking them if they have CCTV cameras in their vicinity.

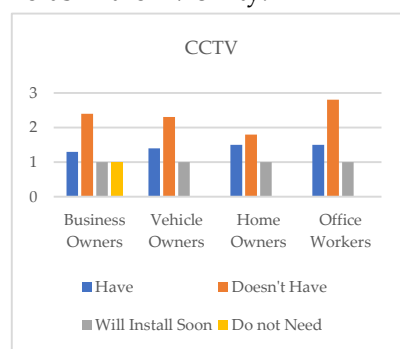


Fig 8: Status of CCTV camera in Bangladesh

Business owners in Bangladesh want a better solution to sort their products, with more than 50% wanting a CCTV camera and 23% wanting only a security guard.

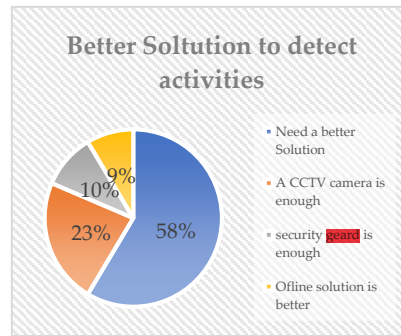


Fig 9: Attitudes of Business Owners to Solution to Detect Activities

Prototype

This prototype will be developed using self-improved data set, CNN, YOLO implemented in a python programming language using NumPy, OpenCV, TensorFlow. As the research is divided into several separate parts, the prototype creation will be done in following steps.

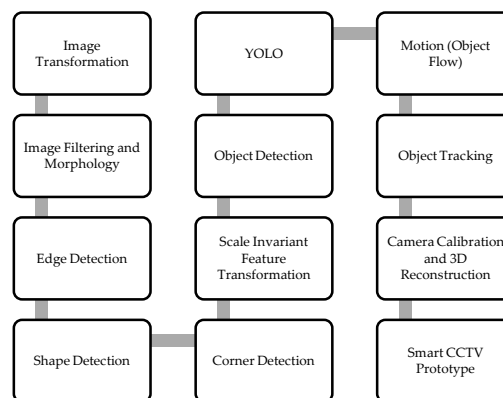


Fig 10: Prototype Overview

Image Transformation is the process of transforming an image into a digital format [17], while Image Filtering and Morphology are the process of blurring and sharpening an image [10], and Edge Detection is the boundary detection [2]. Shape Detection & Hough Transformation are important parts of Image Processing [23], and the Harris Corner Detector algorithm is used to detect general shapes [24]. Scale Invariant Feature Transformation (SIFT) is used for object recognition [25], 3D modelling, video tracking, individual identification of wildlife, and panoramic application [1]. YOLO is an Advanced Object Detector based on Deep Convolutional Neural Network (DCNN) that can be used to detect security issues without time delay and can be used in exam cheating detection.

Dataset

This prototype and project is solely based usage area. Currently the dataset is self-added videos. If restricted video is to be analyzed, then the permission is needed to be given.

System Design

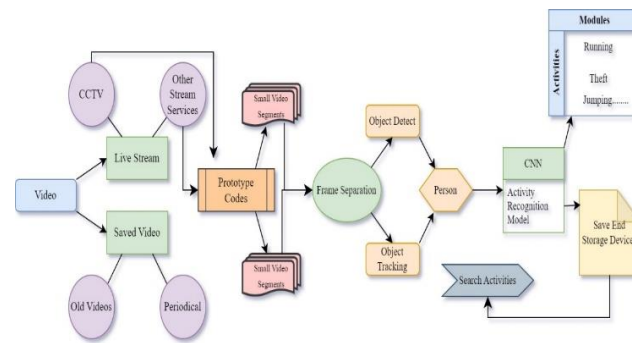


Fig 11: System Design

Evaluation & Result

When we test our model on an abnormal video, we get a video popup and an alert notification. When we test a normal video, we do not send the video to the dashboard and no alert is sent. The training model was analyzed on 5 parameters and 52 videos were used. The model correctly classified 30 videos and gave incorrect output for 22, resulting in an accuracy of 57.7% for 5 epochs.

Table 1: ANALYSIS OF 5 EPOCHS

Activities	Correct	Incorrect	Total
Running	2	7	9
Walking on Roads	7	2	9
Theft	4	4	8
Jumping	7	7	14
Normal	10	2	12
Total	30	22	52

The training model correctly classifies 35 videos out of 52 videos and gives incorrect output for 17, resulting in an accuracy of 67.30% for 15 epochs.

Table 2: ANALYSIS OF 15 EPOCHS

Activities	Correct	Incorrect	Total
Running	5	4	9
Walking on Roads	7	2	9
Theft	5	3	8
Jumping	7	7	14
Normal	11	1	12
Total	35	17	52

The training model correctly classifies 43 videos and gives incorrect output for 9, resulting in an accuracy of 80.76% for 50 epochs.

Table 3: ANALYSIS OF 50 EPOCHS

Activities	Correct	Incorrect	Total
Running	7	2	9
Walking on Roads	8	1	9
Theft	6	2	8

Jumping	8	6	14
Normal	11	1	12
Total	42	10	52

So, if we compare all the approaches we can come with the following conclusion:

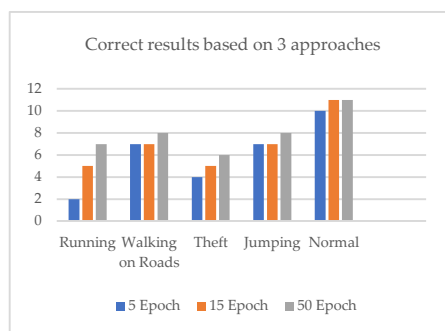


Fig 12: Correct Result Based on 3 Approaches

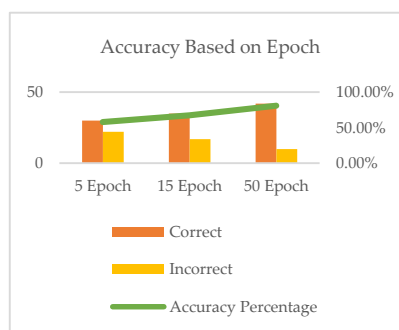


Fig 13: Accuracy Based on Epoch

Conclusion

We attempted a strategy for spotting abnormalities in the supplied footage, such as stealing, persons strolling on the road, leaping, and so on, and instantly notifying the appropriate authorities by email. The data was separated into training and testing in an 80:20 ratio in the first stage, and the foreground data was obtained by subtracting the background data. The model was created, and feature extraction was performed using an auto-encoder architecture. Lastly, aberrant occurrences are discovered using pattern matching. We utilized the self-imported dataset, which is 7 GB in size. The suggested approach has a 5-epoch accuracy of 54.4.7%, a 15-epoch accuracy of 64.4.30% and a 50-epoch accuracy of 80.76%.

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